

# LegalEase AI — Project Structure

Complete folder and file hierarchy

```
LegalEase-AI/
├── frontend/
│   ├── app.py                # Main Streamlit Web Application
│   ├── components/
│   │   ├── input_form.py    # Form with smart contract presets & date pickers
│   │   ├── preview.py       # Full-size document preview & inline editor
│   │   ├── download_buttons.py # Multi-format download streams (.txt, .docx, .pdf)
│   │   └── utils/
│   │       ├── api_client.py # REST client communicating with FastAPI
│   │       └── styles/
│   │           └── custom.css # Unique executive theme stylesheet
│   └──
├── backend/
│   ├── main.py               # FastAPI application entrypoint & middleware
│   ├── config.py             # Environment settings & constants
│   ├── api/
│   │   ├── routes.py         # POST /generate endpoint
│   │   ├── health.py         # GET / and GET /health endpoints
│   │   └── models/
│   │       ├── schemas.py    # Pydantic validation schemas
│   │       └── ai_core/
│   │           ├── gemini_generator.py # Gemini AI orchestration & prompt engineering
│   │           └── services/
│   │               ├── document_service.py # Orchestration & HTML preview generator
│   │               ├── txt_generator.py    # Plain text (.txt) generator
│   │               ├── docx_generator.py   # Microsoft Word (.docx) generator
│   │               └── pdf_generator.py    # ReportLab PDF (.pdf) generator
│   │           └── utils/
│   │               ├── validation.py       # Input and logo validation logic
│   │               ├── sanitization.py     # Text sanitization & terms table parser
│   │               └── error_handler.py    # Custom exceptions & safe error handlers
│   └──
├── assets/
│   └── logo.png              # High-resolution LegalEase brand logo
├── tests/
│   ├── test_health.py        # Health check & root API tests
│   ├── test_generate.py      # POST /generate endpoint tests
│   ├── test_validation.py    # Input & date validation tests
│   └── test_document_generation.py # TXT, DOCX, PDF, and preview tests
├── docs/
│   ├── WORKFLOW.md           # End-to-end workflow documentation
│   ├── SYSTEM_ARCHITECTURE.md # Architecture, component & data flow specs
│   ├── FRONTEND.md           # Streamlit UI & component documentation
│   ├── BACKEND.md            # FastAPI service & backend logic docs
│   ├── API_GATEWAY.md        # REST API endpoints & payload specifications
│   ├── AI_INTEGRATION.md     # Gemini prompt engineering & fallback docs
│   ├── CORE_FUNCTIONALITIES.md # In-depth breakdown of features
│   ├── DEVELOPMENT_SETUP.md  # Step-by-step setup guide for Windows
│   ├── DEPLOYMENT.md         # Local & cloud production deployment guide
│   ├── TESTING.md            # Test suite specifications & results
│   └── CONCLUSION.md         # Project conclusion & future roadmap
├── .env                      # Active environment configuration with API key
├── .env.example              # Environment variables template
├── .gitignore                # Git exclusion rules
├── requirements.txt          # Project dependencies
├── README.md                 # Master project documentation
└── run_project.bat           # Windows one-click batch launcher
```

**Project Stack:** Streamlit Frontend • FastAPI Backend • Gemini AI • TXT/DOCX/PDF Generation • Automated Tests • Documentation